

## PROJECT DOCUMENTATION

# Financial Risk Analysis & Credit Portfolio Optimization

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*AWS RDS • ODBC • Power Query • DAX • Power BI*

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**Tech Stack:** AWS RDS, ODBC Driver, Power Query (ETL), DAX, Power BI

**Target Domain:** Financial Services / Credit Risk Management / Retail Banking

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*End-to-End Cloud Data Engineering & Business Intelligence Solution*

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## 1. Executive Summary

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### 1.1 Project Overview

This project presents an end-to-end data engineering and business intelligence solution created to monitor credit risk, analyze customer borrowing habits, and minimize expected financial losses across a retail lending portfolio. Using a live cloud database on Amazon Web Services (AWS RDS) connected seamlessly to Power BI via an ODBC driver, this project transforms raw transactional data into actionable operational insights.

### 1.2 Core Metrics Summary

| Core Metric                       | Value            |
|-----------------------------------|------------------|
| Total Customer Base Monitored     | 11.81K customers |
| Total Outstanding Debt Exposure   | \$131.21 Million |
| Expected Loss Exposure            | \$11.44 Million  |
| Average Portfolio Interest Rate   | 14.21%           |
| Average Payment Delay (Portfolio) | 20.63 Days       |

## 2. Business Context & Problem Statement

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### 2.1 The Challenge

Lending institutions lose millions of dollars annually due to bad credit underwriting and a failure to spot early warning signs of borrower distress. Static credit scores alone do not reflect real-time customer behavior, leading banks to grant credit lines to individuals who are already over-leveraged across multiple institutions.

### 2.2 Simple Real-World Example

Consider two borrowers applying for a credit expansion:

- Borrower A: Holds 2 bank accounts, maintains a single loan, pays off bills with a minor 7-day average delay, and uses 30% of their credit limit.
- Borrower B: Spreads balances across 9 bank accounts, carries 6 active loans simultaneously, applies for credit constantly, and pays bills nearly 40 days late.

Without an automated system to flag account fragmentation and credit-seeking behavior, standard underwriting tools treat both borrowers similarly until Borrower B completely defaults.

### 2.3 Project Objectives

- Centralize customer risk profiles in a secure cloud database (AWS RDS).
- Build an automated live data connection to Power BI via ODBC.
- Perform comprehensive ETL (Extract, Transform, Load) to clean and structure data.
- Build a 3-tier analytical dashboard covering Executive Risk, Demographic Exposure, and Behavioral Delinquency.
- Provide clear, actionable recommendations to reduce payment delays and protect institutional capital.

### 3. Technology Stack & Data Architecture

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Data flows through the pipeline in the following sequence:

|                    |                    |                 |                   |                    |
|--------------------|--------------------|-----------------|-------------------|--------------------|
| Raw Financial Data | AWS RDS (Cloud DB) | ODBC Connection | Power Query / DAX | Power BI Dashboard |
|--------------------|--------------------|-----------------|-------------------|--------------------|

#### 3.1 Component Breakdown

| Component                              | Role in the Architecture                                                                                                                            |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AWS RDS</b>                         | Central cloud data repository; secure storage, high scalability, ACID-compliant transaction tracking for customer profiles and credit histories.    |
| <b>ODBC Driver</b>                     | Bridges AWS RDS directly into Power BI, enabling live querying without manual file exports or temporary CSV staging.                                |
| <b>Power Query</b>                     | ETL engine inside Power BI; cleans raw inputs, fills null values, normalizes data types, and bins continuous variables into demographic categories. |
| <b>DAX (Data Analysis Expressions)</b> | Powers dynamic calculation measures; computes aggregate metrics such as Total Outstanding Debt, Expected Loss, and Average Delay Days.              |
| <b>Power BI</b>                        | Delivers the interactive visualization interface across three structured reporting pages.                                                           |

## 4. Dataset

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### 4.1 Data Source

The dataset used for this project was sourced from Kaggle, a widely trusted open-data platform for financial and analytics research. It comprises close to 1,00,000 (1 Lakh) individual customer credit records, spanning demographic details, account information, and loan-level attributes. This large sample size provides sufficient statistical depth to identify meaningful patterns in borrower behavior, payment delays, and credit risk segmentation, supporting a reliable, data-driven risk analysis and portfolio optimization model.

## 5. Data Preparation and ETL Process

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### 5.1 Data Extraction

Data was ingested from AWS RDS tables using native SQL queries via the ODBC connection string.

### 5.2 Data Cleaning & Transformation Steps

- Handling Missing Values: Replaced missing values in Credit Mix, Monthly Balance, and Payment Delay fields with calculated statistical medians or default category tags.
- Age Group Categorization: Standardized raw age metrics into 5 intuitive career brackets — Entry Level, Early Career, Mid Career, Established, and Pre-Retirement.
- Income Stratification: Segmented raw annual income figures into 5 socio-economic tiers — Low Income, Lower-Middle, Middle Income, Upper-Middle, and High Income.

### 5.3 Calculated DAX Measures

| Measure                | DAX Formula                                          |
|------------------------|------------------------------------------------------|
| Total Outstanding Debt | SUM ( Outstanding Debt )                             |
| Expected Loss Exposure | SUM ( Potential Loss Amount for High-Risk Accounts ) |
| Average Payment Delay  | AVERAGE ( Delay Dates )                              |

## 6. Data Modeling & Data Ingestion

### 6.1 Data Modeling — Star Schema Design

The data was structured in Power BI as a star schema: a central fact table, `fact_credit_log`, capturing transactional credit behavior (credit score, utilization ratio, payment delay, and investment activity), surrounded by three dimension tables — `dim_customer`, `dim_account`, and `dim_loan` — that describe borrower, account, and loan attributes respectively. `dim_customer` and `dim_account` connect to the fact table in a one-to-many relationship via `customer_id` and `account_id`, while `dim_loan` relates back to `dim_customer` via `customer_id`.

| <code>dim_customer</code>                                                                                            | <code>dim_account</code> | <code>dim_loan</code> |
|----------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------|
| <code>fact_credit_log</code> (central fact table — 1 : many via <code>customer_id</code> / <code>account_id</code> ) |                          |                       |

| Table Name                   | Type            | Key Columns                                                                                                                                                                                                                                                                                  |
|------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fact_credit_log</code> | Fact Table      | <code>log_id</code> , <code>customer_id</code> , <code>account_id</code> , <code>credit_score</code> , <code>credit_history_age</code> , <code>credit_utilization_ratio</code> , <code>delay_from_due_date</code> , <code>amount_invested_monthly</code> , <code>changed_credit_limit</code> |
| <code>dim_customer</code>    | Dimension Table | <code>customer_id</code> , <code>age</code> , <code>annual_income</code> , <code>monthly_inhand_salary</code> , <code>occupation</code>                                                                                                                                                      |
| <code>dim_account</code>     | Dimension Table | <code>account_id</code> , <code>customer_id</code> , <code>credit_mix</code> , <code>interest_rate</code> , <code>num_bank_accounts</code> , <code>num_credit_card</code> , <code>num_of_loan</code>                                                                                         |
| <code>dim_loan</code>        | Dimension Table | <code>customer_id</code> , <code>loan_id</code> , <code>loan_type</code>                                                                                                                                                                                                                     |

### 6.2 Data Ingestion into AWS RDS

The cleaned dataset was uploaded to the AWS RDS instance using DBeaver, a universal SQL client, which was connected to the live AWS RDS database through a secure AWS endpoint connection string. This established the direct pipeline referenced in Section 3, allowing Power BI to draw live data through the ODBC connection in real time.



## 7. Dashboard Analysis & Key Business Findings

### Page 1 — Executive Risk Overview

#### Finding 1 — High Interest Rate Drop-off & Hidden Risk Threshold

**Chart Type:** *Payment Delay based on Interest Rate %*

**Key Finding:** Outstanding debt and late payments spike heavily (peaking above 4K–5K accounts) when interest rates are set between 5% and 20%, but collapse dramatically once interest rates cross 22%.

| ⚠ THE BAD                                                                                                                                                                    | ✓ THE GOOD                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| The bank is heavily exposed in the 5%–20% interest rate bracket, where thousands of borrowers are struggling to pay on time despite carrying standard or low interest rates. | Higher interest rates above 22% naturally filter out low-quality borrowers, leaving a smaller, more disciplined group of accounts. |

**If Left Uncorrected:** Keeping massive debt volumes locked in the 5%–20% rate bucket without stricter payment enforcement will lead to steady profit erosion, as high late-payment volumes in this tier eat away the bank's profit margins over time.

**How to Fix It:** Restructure pricing tiers by tightening credit qualification rules for the 10%–20% interest rate tier and applying small rate surcharges for borrowers in this bucket who delay payments beyond 7 days.

#### Finding 2 — Multi-Account Delay Explosion (Account Fragmentation)

**Chart Type:** *Payment Delay based on holding Bank Accounts*

**Key Finding:** Payment delays stay low (under 10 days) for borrowers holding 0 to 3 bank accounts, but jump to nearly 40 delay days once a borrower holds 8 to 10 bank accounts.

| ⚠ THE BAD                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customers holding 7+ bank accounts are using multiple accounts to shuffle cash, stack debts across institutions, and mask insolvency, causing severe collection delays. |

**If Left Uncorrected:** Sustained 40-day payment delays turn directly into non-performing assets (NPAs). Borrowers will continue debt-cycling between bank accounts until they hit total bankruptcy and default completely with zero recovery yield.

**How to Fix It:** Update the automated underwriting algorithm to automatically cap credit lines or flag high-risk alerts for any applicant operating more than 5 bank accounts.

## Page 2 — Demographic Risk Analysis

### Finding 1 — High Credit Dependence Among Low-Income Borrowers

**Chart Type:** Credit Utilization based on Income Category

**Key Finding:** Credit utilization drops continuously as income increases — Low Income borrowers max out their credit the most at 0.98M, while High Income borrowers use only 0.18M (more than a 5x gap).

| ⚠ THE BAD                                                                                                                                        | ✓ THE GOOD                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Low-income customers are using credit cards and revolving lines to cover basic living costs, keeping them right on the edge of financial stress. | This group produces high interest income and steady card usage if they keep making monthly payments. |

**If Left Uncorrected:** During inflationary periods or price hikes, these low-income borrowers will run out of available credit, causing a sudden spike in unpaid balances and default losses.

**How to Fix It:** Reduce maximum revolving credit limits for low-income profiles and require updated income verification every 6 months to prevent over-borrowing.

### Finding 2 — Mid-Career Credit Score Dominance & Age-Based Drop-off

**Chart Type:** Credit Risk Score based on Age Category

**Key Finding:**

- Total aggregated credit score peaks in the Mid Career group at 29K, followed by Established (23K) and Early Career (20K).
- There is a sharp decline in total score volume moving toward younger borrowers (Entry at 12K) and older borrowers (Pre-Retirement at 10K).

| ⚠ THE BAD                                                                                                                                                                                                                                                         | ✓ THE GOOD                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The bank's credit score capital is heavily concentrated in a single demographic (Mid Career holds nearly 3x the score volume of Pre-Retirement). Any broad financial stress in prime working-age households will impact the portfolio's top-tier credit profiles. | Mid-Career accounts hold the strongest combined credit health (29K aggregate), making them prime candidates for higher-tier lending and cross-selling premium products. |

**If Left Uncorrected:**

- Over-relying on Mid-Career customers while ignoring the growth of Entry (12K) and Early Career (20K) accounts creates a pipeline problem.
- As Mid-Career accounts age into Pre-Retirement (where score volume drops to 10K), the bank will lose its primary revenue drivers without an established pipeline of high-scoring younger borrowers to replace them.

**How to Fix It:**

- Implement loyalty and retention programs for Mid-Career accounts to preserve their 29K score stability.

- Create targeted credit-builder products for Entry (12K) and Early Career (20K) borrowers to nurture their credit scores before they reach their peak earning years.

## Page 3 — Behavioral Delinquency Analysis

### Finding 1 — Income Disparity & Debt Burden Trajectory

**Chart Type:** Debt Burden Trajectory on Income Groups (supported by Income Category Demographics)

**Key Finding:** Total debt load drops significantly as income rises — Low Income and Middle Income categories hold the highest total debt load (peaking near 5K for Poor credit score profiles), whereas High Income and Upper-Middle categories hold significantly lower debt loads (around 2K). Furthermore, Low Income (3.88K) and Lower-Middle (3.23K) make up over 60% of the total customer base.

| ⚠ THE BAD                                                                                                                                                                 | ✓ THE GOOD                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| The bank's debt exposure is concentrated in lower-income segments who carry the heaviest relative debt loads, creating a fragile customer base with limited cash buffers. | Higher-income segments carry lower debt and represent an under-utilized opportunity for safe credit expansion. |

**If Left Uncorrected:** Heavy debt concentration among lower-income groups increases default rates during economic slowdowns, as these households lack the financial savings needed to manage sustained debt burdens.

**How to Fix It:** Re-balance the lending portfolio by offering tailored, low-limit credit products to lower-income groups while expanding credit opportunities for higher-income borrowers.

### Finding 2 — Excessive Credit Inquiries Increase Risk

**Chart Type:** Credit-Seeking Intensity vs. Payment Delays

**Key Finding:** Borrowers with 15–20 payment delays generate an extreme surge in credit inquiries (peaking at 39K inquiries), while carrying an average of 6.36 active loans in the Bad Credit Mix tier.

| ⚠ THE BAD                                                                                             | ✓ THE GOOD                                                                                            |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Signals a “desperation borrowing loop” — taking out new credit simply to pay off older overdue loans. | Inquiry-frequency data can be automated into a real-time hard-stop rule at negligible operating cost. |

**If Left Uncorrected:** Extending credit to desperate borrowers leads directly to 100% loss realization, inflating non-performing asset (NPA) counts.

**How to Fix It:** Enforce an automated hard stop on new credit extensions if a user has more than 10 credit inquiries in a quarter or holds over 4 active loans.

## 8. Project Outcomes & Business Value Delivered

| Outcome                               | Business Value Delivered                                                                                                                                               |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Integrated Cloud Data Pipeline</b> | Established a direct connection from AWS RDS to Power BI via ODBC, removing manual data exports and establishing a single source of truth.                             |
| <b>Loss Prevention</b>                | Identified early warning triggers to help protect \$11.44 Million in Expected Loss Exposure.                                                                           |
| <b>Data-Driven Underwriting</b>       | Shifted underwriting strategy from static credit scores to dynamic behavioral markers (account fragmentation count, credit inquiry frequency, and income utilization). |
| <b>Improved Cash Flow</b>             | Provided clear policy adjustments designed to lower average portfolio payment delays from 20.63 days down to under 12 days.                                            |

***“From Static Credit Scores to Dynamic, Behavior-Driven Risk Intelligence.”***